

SEAN PREVITY

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EDUCATION

University of Central Florida, Computer Science, M.S.	2026 — Present
University of Central Florida, Computer Science, Bachelors of Science	2021 — 2025
• GPA: 3.64	

PROJECTS

Pegasus Parking Website October - December 2024

- Developed a MERN application for coordinating university parking availability.
- Deployed the website to **AWS (EC2)** with HTTPS encryption via Let's Encrypt.
- Built a modular Express.js backend with Mongoose for scalable data modeling and API endpoints.
- Integrated Google Maps Routes API to calculate real-time directions and distances and to find the closest parking location out of 14 possible campus locations.

Babel Translation January — July 2025

- Developed a proof-of-concept translation platform with a React frontend and FastAPI backend.
- Collaborated in an **Agile** team using Jira sprints.
- Researched, tested, and benchmarked open-source AI models with Hugging Face, COMET, and BLEU scores.
- Fine-tuned models to achieve up to **35%** accuracy improvements on multiple low-resource languages.
- Created document parsing and layout-preservation logic for DOCX and PDF files.

France Travel Guide Website April — September 2025

- Designed and deployed a **bilingual** (French/English) travel site using Next.js, Node.js, and PostgreSQL with **37,000** rows of public INSEE data.
- Utilized the OpenAI API for city descriptions and Wikipedia MediaWiki API for photos.
- Implemented MapBox-powered interactive maps for browsing towns by location.
- Reduced API costs and latency by **50%** via an AI description caching system.

European Football Stats App October - December 2025

- Created a full-stack analytics platform for the **top 5 European football leagues**, supporting match previews, historical analysis, and predictive insights.
- Built with **Next.js**, **Node.js**, and **PostgreSQL (Supabase)** with automated data updates.
- Engineered a **Monte Carlo simulation** model using team form, historical performance, and contextual match data to generate probabilistic match outcome predictions.
- Stored data for **200+ teams**, **3000+ players**, **35000+ matches**, and **20+ years** of league standings .
- Implemented **PostgreSQL upsert and conflict-resolution logic** to ensure data consistency across live match updates, odds feeds, and historical records.

SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, C
- **Frameworks/Libraries:** React.js, Next.js, Node.js, Express.js, Redux, Flask, FastAPI, Mongoose, SQLAlchemy, Hugging Face Transformers
- **Databases/Cloud:** PostgreSQL, MySQL, MongoDB, Supabase, AWS, Docker, Drizzle ORM
- **Tools:** Git, GitHub, Postman, SwaggerHub, Jira, Teams
- **Spoken Languages:** English (Native), French (Limited Working)